



Abeda Inamdar Senior College, Pune

T. Y. B.Sc. (Computer Science)

SEMESTER I

2023

Lab Course

Practical course based

On

AngularJS

Assignment Completion Sheet

Lab Course

Practical course based On AngularJS

Sr. No	Assignment Name	Marks (out of 5)	Instruct or Sign
1	AngularJS Services		
2	Assignment based on Directives		
3	Assignment based on different controller		
4	Assignment based on filter		
5	Assignment based on form		
Total (Out of 25)			
Total (Out of 5)			

This is to certify that Mr./Ms. _____

Has successfully completed the **Practical course based On AngularJS** and has

Scored _____ Marks out of 05.

Instructor

H.O. D / Coordinator

Internal Examiner

External Examiner

Assignment 1 AngularJS Services

What is Service?

In AngularJS, a service is a function, or object, that is available for, and limited to, your AngularJS application. In AngularJS you can make your own service, or use one of the many built-in services. AngularJS has about 30 built-in services.

\$location service: The \$location service has methods which return information about the location of the current web page: <pre><html> <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angular.min.js"></script> <body> <div ng-app="myApp" ng-controller="myCtrl"> <p>The url of this page is:</p> <h3>{{myUrl}}</h3> </div> <script> var app = angular.module('myApp', []); app.controller('myCtrl', function(\$scope, \$location) { \$scope.myUrl = \$location.absUrl(); }); </script></body></html></pre>	The \$timeout Service The \$timeout service is AngularJS' version of the window.setTimeout function. <pre><html><script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angular.min.js"></script> <body><div ng-app="myApp" ngcontroller="myCtrl"> <p>This header will change after two seconds:</p> <h1>{{myHeader}}</h1> </div> <p>The \$timeout service runs a function after a specified number of milliseconds.</p> <script> var app = angular.module('myApp', []); app.controller('myCtrl', function(\$scope, \$timeout) { \$scope.myHeader = "Hello World!"; \$timeout(function () { \$scope.myHeader = "How are you today?"; }, 2000); });</script></body></html></pre>
The \$interval Service The \$interval service is AngularJS' version of the window.setInterval function. <pre><html> <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angular.min.js"></script> <body></pre>	Create Your Own Service To create your own service, connect your service to the module: <pre><html> <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angular.min.js"></script> <body></pre>

<pre> <div ng-app="myApp" ng- controller="myCtrl"> <p>The time is:</p> <h1>{{theTime}}</h1> </div> <p>The \$interval service runs a function every specified millisecond.</p> <script> var app = angular.module('myApp', []); app.controller('myCtrl', function(\$scope, \$interval) { \$scope.theTime = new Date().toLocaleTimeString(); \$interval(function () { \$scope.theTime = new Date().toLocaleTimeString(); }, 1000); }); </script></body></html> </pre>	<pre> <div ng-app="myApp" ng- controller="myCtrl"> <p>The hexadecimal value of 255 is:</p> <h1>{{hex}}</h1> </div> <p>A custom service with a method that converts a given number into a hexadecimal number.</p> <script> var app = angular.module('myApp', []); app.service('hexafy', function() { this.myFunc = function (x) { return x.toString(16); } }); app.controller('myCtrl', function(\$scope, hexafy) { \$scope.hex = hexafy.myFunc(257); }); </script></body></html> </pre>
The AngularJS \$http service makes a request to the server, and returns a response. <pre> <html> <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angular.min.js"></script> <body> <div ng-app="myApp" ng-controller="myCtrl"> <p>Today's welcome message is:</p> <h1>{{myWelcome}}</h1> </div> <script> var app = angular.module('myApp', []); app.controller('myCtrl', function(\$scope, \$http) { \$http.get("/welcome.htm").then(function (response) { \$scope.myWelcome = response.data; }); }); </script></body></html> </pre>	

Practice Problems:

- 1) Create angular JS Application that shows location of the current web page.
- 2) Create angular JS Application that demonstrate \$timeout Service

SET A:

- 1) Create angular JS Application that show the current Data and Time of the System (Use Interval Service).
- 2) Create a angular JS Application that greet the User (Use \$timeout Service).

- 3) Using angular js create a SPA to carry out validation for a username entered in textbox. If the textbox is blank, alert 'Enter username'. If the number of characters is less than three, alert 'Username is too short'. If value entered is appropriate the print 'Valid username' and password should be minimum 8 characters.

SET B:

- 1) Using angular js create a SPA to fetch suggestions when if user is typing in a textbox. (eg like google suggestions. Hint create array of suggestions and matching string will be displayed).
- 2) Create angular JS Application that shows location of the current web page.

Assignment Evaluation

0: Not Done []	1: Incomplete []	2: Late Complete []
3: Need Improvement []	4: Complete []	5: Well Done []

Signature of Instructor

Assignment 2: based on Directives

Example 1: Simple AngularJS Script to add two numbers.

```
<html>

<script
src="http://ajax.googleapis.com/ajax/libs/angularjs/1.4.8/angular.
min.js"></script>

<body>

<div ng-app>

<p>A simple expression example: {{ 5 + 5 }}</p>
```

AngularJS directives are used to extend HTML. They are special attributes starting with ngprefix. Let us discuss Some Common directives –

- **ng-app** – This directive starts an AngularJS Application.
- **ng-init** – This directive initializes application data.
- **ng-model** – This directive defines the model that is variable to be used in AngularJS.
- **ng-repeat** – This directive repeats HTML elements for each item in a collection.

ng-app directive

It defines the root element. It automatically initializes or bootstraps the application when the web page containing AngularJS Application is loaded. It is also used to load various AngularJS modules in AngularJS Application.

```
<div ng-app = "">
```

```
...
```

```
</div>
```

ng-init directive

The ng-init directive initializes an AngularJS Application data. It is used to assign values to the variables.

ng-model directive

The ng-model directive defines the model/variable to be used in AngularJS Application. In the following example, we define a model named name.

```
<div ng-app = "">  
  <p>Enter your Name: <input type = "text" ng-model = "name"></p>  
</div>
```

ng-repeat directive

The ng-repeat directive repeats HTML elements for each item in a collection.

Example

The following example shows the use of all the above-mentioned directives.

```
<!DOCTYPE html>  
<html>  
  <script src="angular.min.js"></script>  
  <body>  
    <div ng-app="" ng-init="firstName='Sara'">  
      <p>Input something in the input box:</p>  
      <p>Name: <input type="text" ng-model="firstName"></p>  
      <p>You wrote: {{ firstName }}</p>  
    </div>  
  </body>  
</html>
```

AngularJS Directives List

Directives	Description
ng-app	It defines the root element of an application.
ng-bind	It binds the content of an html element to application data.
ng-bind-html	It binds the inner HTML of an HTML element to application data, and also removes dangerous code from the html string.
ng-bind-template	It specifies that the text content should be replaced with a template.
ng-blur	It specifies a behavior on blur events.
ng-change	It specifies an expression to evaluate when content is being changed by the user.
ng-checked	It specifies if an element is checked or not.
ng-class	It specifies css classes on html elements.
ng-class-even	It is same as ng-class, but will only take effect on even rows.
ng-class-odd	It is same as ng-class, but will only take effect on odd rows.
ng-click	It specifies an expression to evaluate when an element is being clicked.
ng-cloak	It prevents flickering when your application is being loaded.
ng-controller	It defines the controller object for an application.
ng-copy	It specifies a behavior on copy events.
ng-csp	It changes the content security policy.
ng-cut	It specifies a behavior on cut events.
ng-dblclick	It specifies a behavior on double-click events.
ng-focus	It specifies a behavior on focus events.

ng-hide	It hides or shows html elements.
ng-href	It specifies a URL for the <a> element.
ng-if	It removes the html element if a condition is false.
ng-include	It includes html in an application.
ng-init	It defines initial values for an application.
ng-jq	It specifies that the application must use a library, like jQuery.
ng-keydown	It specifies a behavior on keydown events.
ng-keypress	It specifies a behavior on keypress events.
ng-keyup	It specifies a behavior on keyup events.
ng-list	It converts text into a list (array).
ng-open	It specifies the open attribute of an element.
ng-options	It specifies <options> in a <select> list.
ng-paste	It specifies a behaviour on paste events.
ng-pluralize	It specifies a message to display according to en-us localization rules.
ng-readonly	It specifies the read-only attribute of an element.
ng-required	It specifies the required attribute of an element.
ng-selected	It specifies the selected attribute of an element.
ng-show	It shows or hides html elements.
ng-src	It specifies the src attribute for the element.
ng-srcset	It specifies the srcset attribute for the element.
ng-style	It specifies the style attribute for an element.
ng-submit	It specifies expressions to run on onsubmit events.
ng-switch	It specifies a condition that will be used to show/hide child elements.

AngularJS has its own HTML events directives.

ng-blur, ng-change, ng-click, ng-copy , ng-cut, ng-dblclick, ng-focus, ng-keydown, ngkeypress ng-keyup, ng-mousedown, ng-mouseenter, ng-mouseleave, ng-mousemove, ngmouseover ,ng-mouseup, ng-paste and so...on

Example 1 : ng-click Event

```
<body ng-app="myApp">
<div ng-controller="myCtrl">
  <button ng-click="myFunc()">OK</button>
  <p>The button has been clicked {{count}} times.</p>
</div>
<script>
angular.module('myApp', [])
.controller('myCtrl', ['$scope', function($scope) {
  $scope.count = 0;
  $scope.myFunc = function() {
    $scope.count++;
  };
}]);
</script>
</body>
```

Practice Programs:

- 1) Write angular JS Program to bind product of 2 numbers.
- 2) Write angular JS Program to perform Arithmetic operation on two number
- 3) Write an angular JS Program to count movement of the cursor on the screen

SET A:

- 1) Write an AngularJS script for addition of two numbers using ng-init, ng-model & ng-bind. And also demonstrate ng-show, ng-disabled, ng-click directives on button component.
- 2) Write an AngularJS script to display list of games stored in an array on click of button using ng-click. And also demonstrate ng-init, ng-bind directive of Angularjs.
- 3) create a simple directive that changes the background color of an element when it's clicked.

SET B :

- 1) Write angular JS by using ng-click Directive to display an alert message after clicking the element.
- 2) Write an AngularJS script to display Student name, Roll no and calculated Percentage.(use ng-init to initialize name,roll and all subject marks).
- 3) Write an AngularJS script to display Product name, Quantity, Rate and Total Price.(use ng-init to initialize values).

Assignment Evaluation

0: Not Done []	1: Incomplete []	2: Late Complete []
3: Need Improvement []	4: Complete []	5: Well Done []

Signature of Instructor

Assignment 3: based on different controller

What is Controller?

AngularJS controllers are used to control the flow of data of AngularJS application. A controller is defined using ng-controller directive. A controller is a JavaScript object containing attributes/properties and functions. Each controller accepts \$scope as a parameter which refers to the application/module that controller is to control.

```
<html>
<script
src="http://ajax.googleapis.com/ajax/libs/angularjs/
1.4.8/angular.min.js">
</script>
<body>
<div ng-app="myApp" ng-controller="myCtrl">
First Name: <input type="text" ng-
model="firstName"><br>
Last Name: <input type="text" ng-
model="lastName"><br>
<br>
Full Name: {{fullName()}} <!--Function Calling --
>
</div>
<script> var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope) {
$scope.firstName = "Satish";
$scope.lastName = "Mulgi";
$scope.fullName = function() {
return $scope.firstName + " " + $scope.lastName;
};
});
10. </script> </body> </html>
```

Here, the AngularJS application runs inside the <div> is defined by ng-app="myApp". and the AngularJS directive is ng-controller="myCtrl" attribute. The myCtrl function is a JavaScript function. AngularJS will invoke the controller with a \$scope object. The controller creates two properties (variables) in the scope (firstName and lastName). The ng-model directives bind the input fields to the controller properties (firstName and lastName).

- 1) View, which is the HTML.
- 2) Model, which is the data available for the current view.
- 3) Controller, which is the JavaScript function that makes/changes/removes/controls the data.

SET A:

1. Using angular js create a SPA(Single Page Application) to display the details of product (srNo,Name, Price, quantity) available in the departmental stores.
2. Create Simple AngularJS Script to Print Student Marks card with grade using controller.

SET B:

- 1) Using Angular JS Create a SPA for Bus Ticket Reservation consisting of fields : Name, Address, contact no, source station(Dropdown list), Destination station(Dropdown list),Date of booking, date of journey, name of passenger, gender of passenger etc. Display the e –Ticket.
- 2) Using Angular JS Create a SPA to take the information of a customer for booking Ticket consisting of fields such as name, address, contact no., gender, Date of booking, date of journey, name of passengers etc. Display the e –Ticket.

Assignment Evaluation

- | | | |
|-------------------------|-------------------|----------------------|
| 0: Not Done [] | 1: Incomplete [] | 2: Late Complete [] |
| 3: Need Improvement [] | 4: Complete [] | 5: Well Done [] |

Signature of Instructor

Assignment 4: based on Filter

AngularJS filters are used to format data, Following is a list of filters used for transforming data.

currency: Format a number to a currency format.
date : Format a date to a specified format.
filter : Select a subset of items from an array. **limitTo :** Limits an array/string, into a specified number of elements/characters.

lowercase : Format a string to lower case.
number : Format a number to a string.
orderBy : Orders an array by an expression.
uppercase : Format a string to upper case.
json: Format an object to a JSON string

Adding Filters to Expressions Filters can be added to expressions by using the pipe character |, followed by a filter.

Example 1 : The uppercase filter format strings to upper

```
<div ng-app="myApp" ng-controller="personCtrl">
  <p>The name is {{ lastName | uppercase }}</p>
</div>
<script>
var app = angular.module('myApp', []);
app.controller('personCtrl', function($scope) {
  $scope.lastName = "Satish Mulgi";
});
</script>
```

Example 2 : The lowercase filter format strings to lower

```
<div ng-app="myApp" ng-controller="personCtrl">
  <p>The name is {{ lastName | lowercase }}</p>
</div>
<script>
var app = angular.module('myApp', []);
app.controller('personCtrl', function($scope) {
  $scope.lastName = "Satish Mulgi";
});
</script>
```

SET A:

1. Store 10 students information in an array and display students information in table form orderBy Name (use orderBy filter sorts an array).
2. Using angular js display the student details who are live in pune in Table format (using ng-repeat directive, use Array to store data, use filter).
3. Write an AngularJS program that show the use of filters to filter a list of items based on user input

SET B:

1. Write an AngularJS Script to search a product with its rate (use ng-repeat directive, use Array to store data, use filter currency).
2. Write an AngularJS script to search student name according to the character typed and display details (use array and filter).
3. Using angular js display the Employee details order by salary in Table format (using ng-repeat directive, use Array to store data, use filter)

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement [] 4: Complete []

5: Well Done []

Signature of Instructor

Assignment 5: based on forms

Form Validation

AngularJS offers client-side form validation. AngularJS monitors the state of the form and input fields (input, textarea, select), and lets you notify the user about the current state. AngularJS also holds information about whether they have been touched, or modified, or not. we can use standard HTML5 attributes to validate input, or you can make your own validation functions.

Form State and Input State AngularJS is constantly updating the state of both the form and the input fields. Input fields have the following states: • \$untouched : The field has not been touched yet • \$touched : The field has been touched • \$pristine : The field has not been modified yet • \$dirty : The field has been modified • \$invalid : The field content is not valid • \$valid : The field content is valid They are all properties of the input field, and are either true or false.	Forms have the following states: • \$pristine : No fields have been modified yet • \$dirty : One or more have been modified • \$invalid : The form content is not valid • \$valid : The form content is valid • \$submitted : The form is submitted They are all properties of the form, and are either true or false. You can use these states to show meaningful messages to the user. Example, if a field is required, and the user leaves it blank, you should give the user a warning:
Required <pre><html> <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angular.min.js"></script> <body ng-app=""> <p>Try leaving the first input field blank:</p> <form name="myForm"> <p>Name: <input name="myName" ng-model="myName" required> The name is required. </p> <p>Address: <input name="myAddress" ng-model="myAddress" required> </p> </form> <p>We use the ng-show directive to only show the error message if the field has been touched AND is empty.</p></body> </html></pre>	

SET A:

- 1) Using AngularJS Create a simple registration form that collects the user's name, email, and password. We'll also implement form validation to ensure that all fields are filled correctly.
- 2) Write an AngularJS Program to create student registration form and validate all fields.
- 3) Write an AngularJS program to create a simple form using AngularJS form directives and handling form submission.

SET B:

- 1) Using angular js create a SPA that to accept the details such as name, mobile number, and pin-code and email address and make validation. Name should contain character only, mobile number should contain only 10 digit, Pin code should contain only 6 digit, email id should contain only one @, Symbol.
- 2) Write an AngularJS Program to create a SPA (Single Page application) for Login System.

Assignment Evaluation

0: Not Done []	1: Incomplete []	2: Late Complete []
3: Need Improvement []	4: Complete []	5: Well Done []

Signature of Instructor